

Chapter 7

Quantile Regression

Here's a prayer for you. Got a pencil? . . . 'Protect me from knowing what I don't need to know. Protect me from even knowing that there are things to know that I don't know. Protect me from knowing that I decided not to know about the things I decided not to know about. Amen.' There's another prayer that goes with it. 'Lord, lord, lord. Protect me from the consequences of the above prayer.'

Douglas Adams, *Mostly Harmless* (1995)

Rightly or wrongly, 95 percent of applied econometrics is concerned with averages. If, for example, a training program raises average earnings enough to offset the costs, we are happy. The focus on averages is partly because obtaining a good estimate of the average causal effect is hard enough. And if the dependent variable is a dummy for something like employment, the mean describes the entire distribution. But many variables, like earnings and test scores, have continuous distributions. These distributions can change in ways not revealed by an examination of averages, for example, they can spread out or become more compressed. Applied economists increasingly want to know what's happening to an entire distribution, to the relative winners and losers, as well as to averages.

Policy-makers and labor economists have been especially concerned with changes in the wage distribution. We know, for example, that flat average real wages are only a small part of what's been going on in the labor market for the past 25 years. Upper earnings quantiles have been increasing, while lower quantiles have been falling. In other words, the rich are getting richer and the poor are getting poorer. But that's not all - recently, inequality has grown asymmetrically; for example, among college graduates, it's mostly the rich getting richer, with wages at the lower decile unchanging. The complete story of the changing wage distribution is fairly complicated and would seem to be hard to summarize.

Quantile regression is a powerful tool that makes the task of modeling distributions easy, even when the underlying story is complex and multi-dimensional. We can use this tool to see whether participation in a training program or membership in a labor union affects earnings inequality as well as average earnings. We

can also check for interactions, like whether and how the relation between schooling and inequality has been changing over time. Quantile regression works very much like conventional regression: confounding factors can be held fixed by including covariates; interaction terms work the same as with regular regression, too. And sometimes we can even use instrumental variables methods to estimate causal effects on quantiles when a selection-on-observables story seems implausible.

7.1 The Quantile Regression Model

The starting point for quantile regression is the conditional quantile function (CQF). Suppose we are interested in the distribution of a continuously-distributed random variable, Y_i , with a well-behaved density (no gaps or spikes). Then the CQF at quantile τ given a vector of regressors, x_i , can be defined as:

$$Q_\tau(Y_i|X_i) = F_Y^{-1}(\tau|X_i)$$

where $F_Y(y|X_i)$ is the distribution function for Y_i conditional on X_i . When $\tau = .10$, for example, $Q_\tau(Y_i|X_i)$ describes the lower decile of Y_i given X_i , while $\tau = .5$ gives us the conditional median.¹ By looking at changes in the CQF of earnings as a function of education, we can tell whether the dispersion in earnings goes up or down with schooling. By looking at changes in the CQF of earnings as a function of education and time, we can tell whether the relationship between schooling and inequality is changing over time.

The CQF is the conditional-quantile version of the CEF. Recall that the CEF can be derived as the solution to a mean-squared error prediction problem,

$$E[Y_i|X_i] = \arg \min_{m(X_i)} E[(Y_i - m(X_i))^2].$$

In the same spirit, the CQF solves the following minimization problem,

$$Q_\tau(Y_i|X_i) = \arg \min_{q(X)} E[\rho_\tau(Y_i - q(X_i))], \quad (7.1.1)$$

where $\rho_\tau(u) = (\tau - 1(u \leq 0))u$ is called the "check function" because it looks like a check-mark when you plot it. If $\tau = .5$, this becomes least absolute deviations because $\rho_{.5}(u) = \frac{1}{2}(\text{sign } u)u = \frac{1}{2}|u|$. In this case, $Q_\tau(Y_i|X_i)$ is the conditional median since the conditional median minimizes absolute deviations. Otherwise,

¹More generally, we can define the CQF for discrete random variables and random variables with less-than-well-behaved densities as

$$Q_\tau(Y_i|X_i) = \inf \{y : F_Y(y|X_i) \geq \tau\}.$$

the check function weights positive and negative terms asymmetrically:

$$\rho_{\tau}(u) = 1(u > 0) \cdot \tau u + 1(u \leq 0) \cdot (1 - \tau)u.$$

This asymmetric weighting generates a minimand that picks out conditional quantiles (a fact that's not immediately obvious but can be proved with a little work; see Koenker, 2005).

As a practical tool, the CQF shares the disadvantages of the CEF with continuous or high-dimensional X_i : it may be hard to estimate and summarize. We'd therefore like to boil this function down to a small set of numbers, one for each element of X_i . Quantile regression accomplishes this by substituting a linear model for $q(X_i)$ in (7.1.1), producing

$$\beta_{\tau} \equiv \arg \min_{b \in \mathbb{R}^d} E [\rho_{\tau}(y_i - X_i' b)]. \quad (7.1.2)$$

The quantile regression *estimator*, $\hat{\beta}_{\tau}$, is the sample analog of (7.1.2). It turns out this is a linear programming problem that is fairly easy (for computers) to solve.

Just as OLS fits a linear model to y_i by minimizing expected squared error, quantile regression fits a linear model to y_i using the asymmetric loss function, $\rho_{\tau}(\cdot)$. If $Q_{\tau}(y_i|X_i)$ is in fact linear, the quantile regression minimand will find it (just as if the CEF is linear, OLS will find it). The original quantile regression model, introduced by Koenker and Bassett (1978), was motivated by the assumption that the CQF is linear. As it turns out, however, the assumption of a linear CQF is unnecessary - quantile regression is useful whether or not we believe this.

Before turning to a more general theoretical discussion of quantile regression, we illustrate the use of this tool to study the wage distribution. The motivation for the use of quantile regression to look at the wage distribution comes from labor economists' interest in the question of how inequality varies conditional on covariates like education and experience (see, e.g., Buchinsky, 1994). The overall gap in earnings by schooling group (e.g., the college/high-school differential) grew considerably in the 1980s and 1990s. Less clear, however, is how the wage distribution has been changing *within* education and experience groups. Many labor economists believe that increases in so-called "within-group inequality" provide especially strong evidence of fundamental changes in the labor market, not easily accounted for by changes in institutional features like the percent of workers who belong to labor unions.

Table 7.1.1: Quantile regression coefficients for schooling in the 1970, 1980, and 2000 Censuses

Census	Obs.	Desc. Stats.		Quantile Regression Estimates					OLS Estimates	
		Mean	SD	0.1	0.25	0.5	0.75	0.9	Coeff.	Root MSE
1980	65023	6.4	0.67	.074 (.002)	.074 (.001)	.068 (.001)	.070 (.001)	.079 (.001)	.072 (.001)	0.63
1990	86785	6.46	0.06	.112 (.003)	.110 (.001)	.106 (.001)	.111 (.001)	.137 (.003)	.114 (.001)	0.64
2000	97397	6.5	0.75	.092 (.002)	.105 (.001)	.111 (.001)	.120 (.001)	.157 (.004)	.114 (.001)	0.69

Notes: Adapted from Angrist, Chernozhukov, and Fernandez-Val (2006). The tables reports quantile regression estimates of the returns to schooling, with OLS estimates shown at the right for comparison. The sample includes US-born white and black men aged 40-49. Standard errors are reported in parentheses. All models control for race and potential experience. Sampling weights were used for the 2000 Census estimates.

Table 7.1.1 reports schooling coefficients from quantile regressions estimated using the 1980, 1990, and 2000 Censuses. The models used to construct these estimates control for race and a quadratic function of potential labor market experience (defined to be $age - education - 6$). The .5 quantile coefficients - for the conditional median - are very much like the OLS coefficients at the far right-hand side of the table. For example, the OLS estimate of .072 in the 1980 census is not very different from the .5 quantile coefficient of about .068 in the same data. If the conditional-on-covariates distribution of log wages is symmetric, so that the conditional median equals the conditional mean, we should expect these two coefficients to be the same. Also noteworthy is that fact that the quantile coefficients are similar across quantiles in 1980. An additional year of schooling raises median wages by 6.8 percent, with slightly higher effects on the lower and upper quartiles of .074 and .070. Although the estimated returns to schooling increased sharply between 1980 and 1990 (up to .106 at the median, with an OLS return of .114 percent), there is a reasonably stable pattern of returns across quantiles in the 1990 Census. The largest effect is on the upper decile, a coefficient of .137, while the other quantile coefficients are around .11.

We should expect to see constant coefficients across quantiles if the effect of schooling on wages amounts to what is sometimes called a "location shift." Here, that means that as higher schooling levels raise average earnings, other parts of the wage distribution move in tandem (i.e., within-group inequality does not change). Suppose, for example, that log wages can be described by a classical linear regression model:

$$Y_i \sim N(X_i'\beta, \sigma_\varepsilon^2), \quad (7.1.3)$$

where $E[Y_i|X_i] = X_i'\beta$ and $Y_i - X_i'\beta \equiv \varepsilon_i$ is a Normally distributed error with constant variance σ_ε^2 . Homoskedasticity means the conditional distribution of log wages is no more spread out for college graduates than for high school graduates. The implications of the linear homoskedastic model for quantiles are apparent from the fact that

$$P[Y_i - X_i'\beta < \sigma_\varepsilon \Phi^{-1}(\tau) | X_i] = \tau,$$

where $\Phi^{-1}(\tau)$ is the inverse of the standard Normal CDF. From this we conclude that $Q_\tau(Y_i|X_i) = X_i'\beta + \sigma_\varepsilon \Phi^{-1}(\tau)$. In other words, apart from the changing intercept, all quantile regression coefficients are the same. The results in Table 7.1.1 for 1980 and 1990 are not too far from this stylized representation.

In contrast with the simple pattern in 1980 and 1990 Census data, quantile regression estimates from the 2000 Census differ markedly across quantiles, especially in the right tail. An additional year of schooling raises the lower decile of wages by 9.2 percent and the median by 11.1 percent. In contrast, an additional year of schooling raises the upper decile by 15.7 percent. Thus, in addition to increases in overall inequality in the 1980s and 1990s (a fact we know from simple descriptive statistics), by 2000, inequality began to increase with education as well. This development is the subject of considerable discussion among labor economists, who are particularly concerned with whether it points to fundamental or institutional changes

in the labor market (see, e.g., Autor, Katz, and Kearney (2005) and Lemieux (2008)).

Again, a parametric example helps us see where an increasing pattern of quantile regression coefficients might come from. We can generate increasing quantile regression coefficients by adding heteroskedasticity to the classical Normal regression model, (7.1.3). Suppose that

$$Y_i \sim N(X_i' \beta, \sigma^2(X_i)),$$

where $\sigma^2(X_i) = (\lambda' X_i)^2$ and λ is a vector of positive coefficients such that $\lambda' X_i > 0$ (perhaps proportional to β , so that the conditional variance grows with the conditional mean).² Then

$$P[Y_i - X_i' \beta < (\lambda' X_i) \Phi^{-1}(\tau) | X_i] = \tau,$$

with the implication that

$$Q_\tau(Y_i | X_i) = X_i' \beta + (\lambda' X_i) \Phi^{-1}(\tau) = X_i' [\beta + \lambda \Phi^{-1}(\tau)]. \quad (7.1.4)$$

so that quantile regression coefficients increase across quantiles.

Putting the pieces together, Table 7.1.1 neatly summarizes two stories, both related to variation in within-group inequality. First, results from the 2000 Census show inequality increasing sharply with education. The increase is asymmetric, however, and appears much more clearly in the upper tail of the wage distribution. Second, this increase is a new development. In 1980 and 1990, in contrast, schooling affected the wage distribution in a manner roughly consistent with a simple location shift.³

7.1.1 Censored Quantile Regression

Quantile regression allows us to look at features of the conditional distribution of Y_i when part of the distribution is hidden. Suppose you have data of the form

$$Y_{i,obs} = Y_i \cdot 1[Y_i < c], \quad (7.1.5)$$

²See Card and Lemieux (1996) for an empirical example of a regression model with this sort of heteroskedasticity. Koenker and Portnoy (1996) call this a linear location-scale model.

³The results in table 7.1.1 include two sets of standard errors. The first are conventional standard errors, of the sort reported by Stata's `qreg` command (also specifying "robust"). These presume the CQF is truly linear. The formula for these is

$$\tau(1-\tau)\{E[f_{u_\tau}(0|X_i)X_iX_i']^{-1}E[X_iX_i']E[f_{u_\tau}(0|X_i)X_iX_i']^{-1},$$

where $f_{u_\tau}(0|X_i)$ is the conditional density of the quantile-regression residual at zero. If the residuals are homoskedastic this simplifies to $\frac{\tau(1-\tau)}{f_{u_\tau}^2(0)}E[X_iX_i']^{-1}$. The second set are robust to misspecification, computed using formulas in Angrist, Chernozhukov, and Fernandez-Val (2006). In this example, the impact of nonlinearity on standard errors is minor.

where $Y_{i,obs}$ is what you get to see and Y_i is the variable you would like to see. The variable $Y_{i,obs}$ is *censored* - information about Y_i in $Y_{i,obs}$ is limited for confidentiality reasons or because it was too difficult or time-consuming to collect more information. In the CPS, for example, high earnings are topcoded to protect respondent confidentiality. This means data above the topcode are recoded to have the topcode value. Duration data may also be censored: in a study of the effects of unemployment insurance on the duration of employment, we might follow new UI claimants for up to 40 weeks. Anyone out of work for longer has an unemployment spell length that is censored at 40. Note that limited dependent variables like hours worked or medical expenditure, discussed in Section 3.4.2, are not censored; they commonly take on the value zero by their nature, just as dummy variables like employment status do.

When dealing with censored dependent variables, quantile regression can be used to estimate the effect of covariates on conditional quantiles that are below the censoring point (assuming censoring is from above). This reflects the fact that recoding earnings above the upper decile to be equal to the upper decile has no effect on the median. So if CPS topcoding affects relatively few people (as is often true), censoring has no effect on estimates of the conditional median or even β_τ for $\tau = .75$. Likewise, if less than 10 percent of the sample is censored conditional on all values of X_i , then when estimating β_τ for τ up to .9 you can simply ignore it. Alternately, you can limit the sample to values of X_i where $Q_\tau(Y_i|X_i)$ is below c (or above, if censoring is from the bottom with $Y_{i,obs} = Y_i \cdot 1[Y_i > c]$).

Powell (1986) formalizes this idea with the censored quantile regression estimator. Because we may not know which conditional quantiles are below the censoring point (continuing to think of top codes), Powell proposes we work with

$$Q_\tau(Y_i|X_i) = \min(c, X_i' \beta_\tau^c).$$

The parameter vector β_τ^c solves

$$\beta_\tau^c \equiv \arg \min_{b \in \mathbb{R}^d} E\{1[X_i' b < c] \cdot \rho_\tau(Y_i - X_i' b)\}. \quad (7.1.6)$$

In other words, we solve the quantile regression minimization problem for values of X_i such that $X_i' \beta_\tau^c < c$. That is, we minimize the sample analog of (7.1.6). As long as there is enough uncensored data, the resulting estimates give us the quantile regression function we would have gotten had the data not been censored (assuming the conditional quantile function is, in fact, linear). And if it turns out that the conditional quantiles you are estimating are below the censoring point, then you are back to regular quantile regression.

The sample analog of (7.1.6) is no longer a linear programming problem but Buchinsky (1994) proposes a simple iterated linear programming algorithm that appears to work well. The iterations go like this: First you estimate β_τ^c ignoring the censoring. Then find the cells with $X_i' \beta_\tau^c < c$. Then estimate the quantile regression again using these cells only, and so on. This algorithm is not guaranteed to converge but it appears to do so in practice. Standard errors can be bootstrapped. Buchinsky (1994) and Chamberlain

(1994) use this approach to estimate the returns to schooling for highly-experienced workers that may have earnings above the CPS topcode. The censoring adjustment tends to increase the returns to schooling for this group.

7.1.2 The Quantile Regression Approximation Property★

The CQF of log wages given schooling is unlikely to be exactly linear, so the assumptions of the original quantile regression model fail to hold in this example. Luckily, quantile regression can also be understood as giving a MMSE linear approximation to the CQF, though in this case the MMSE problem is a little more complicated and harder to derive than for the regression-CEF theorem. For any quantile index $\tau \in (0, 1)$, define the quantile regression specification error as:

$$\Delta_\tau(\mathbf{X}_i, \beta_\tau) \equiv \mathbf{X}_i' \beta_\tau - Q_\tau(Y_i | \mathbf{X}_i).$$

The population quantile regression vector can be shown to minimize an expected weighted average of the squared specification error, $\Delta_\tau^2(\mathbf{X}_i, \beta)$, as shown in the following theorem from Angrist, Chernozhukov, and Fernandez-Val (2006):

Theorem 7.1.1 (Quantile Regression Approximation) *Suppose that (i) the conditional density $f_Y(y|\mathbf{X}_i)$ exists almost surely, (ii) $E[Y_i]$, $E[Q_\tau(Y_i|\mathbf{X}_i)]$, and $E\|\mathbf{X}_i\|$ are finite, and (iii) β_τ uniquely solves (7.1.2). Then*

$$\beta_\tau = \arg \min_{b \in \mathbb{R}^d} E [w_\tau(\mathbf{X}_i, b) \cdot \Delta_\tau^2(\mathbf{X}_i, b)], \quad (7.1.7)$$

where

$$\begin{aligned} w_\tau(\mathbf{X}_i, b) &= \int_0^1 (1-u) \cdot f_{\epsilon(\tau)}(u\Delta_\tau(\mathbf{X}_i, b) | \mathbf{X}_i) du \\ &= \int_0^1 (1-u) \cdot f_Y(u \cdot \mathbf{X}_i' b + (1-u) \cdot Q_\tau(Y_i | \mathbf{X}_i) | \mathbf{X}_i) du \geq 0 \end{aligned}$$

and $\epsilon_i(\tau)$ is a quantile-specific residual,

$$\epsilon_i(\tau) \equiv Y_i - Q_\tau(Y_i | \mathbf{X}_i),$$

with conditional density $f_{\epsilon(\tau)}(e | \mathbf{X}_i)$ at $\epsilon_i(\tau) = e$. Moreover, when Y_i has a smooth conditional density, we have for β in the neighborhood of β_τ :

$$w_\tau(\mathbf{X}_i, \beta) \approx 1/2 \cdot f_Y(Q_\tau(Y_i | \mathbf{X}_i) | \mathbf{X}_i). \quad (7.1.8)$$

The quantile regression approximation theorem looks complicated but the big picture is simple. We can think of quantile regression as approximating $Q_\tau(Y_i|X_i)$, just as OLS approximates $E[Y_i|X_i]$. The OLS weighting function is the histogram of X_i , which we denote $\pi(X_i)$. The quantile regression weighting function, implicitly given by $w_\tau(X_i, \beta_\tau) \cdot \pi(X_i)$, is more elaborate than $\pi(X_i)$ alone (the histogram is implicitly part of the quantile regression weighting function because the expectation in (7.1.7) is over the distribution of X_i). The term $w_\tau(X_i, \beta_\tau)$ involves the quantile regression vector, β_τ , but can be rewritten with β_τ partialled out so that it is a function of X_i only (see Angrist, Chernozhukov, and Fernandez-Val, 2006, for details). In any case, the quantile regression weights are approximately proportional to the density of Y_i in the neighborhood of the CQF.

The quantile regression approximation property is illustrated in Figure 7.1.1, which plots the conditional quantile function of log wages given highest grade completed using 1980 Census data. Here we take advantage of the discreteness of schooling and large census samples to estimate the CQF non-parametrically by computing the quantile of wages for each schooling level. Panels A-C plot a nonparametric estimate of $Q_\tau(Y_i|X_i)$ along with the linear quantile regression fit for the 0.10, 0.50, and 0.90 quantiles, where X_i includes only the schooling variable and a constant. The nonparametric cell-by-cell estimate of the CQF is plotted with circles in the figure, while the quantile regression line is solid. The figure shows how linear quantile regression approximates the CQF.

It's also interesting to compare quantile regression to a histogram-weighted fit to the CQF, similar to that provided by OLS for the CEF. The histogram-weighted approach to quantile regression was proposed by Chamberlain (1994). The Chamberlain minimum distance (MD) estimator is the sample analog of the vector $\tilde{\beta}_\tau$ obtained by solving

$$\tilde{\beta}_\tau = \arg \min_{b \in \mathbb{R}^d} E[(Q_\tau(Y_i|X_i) - X_i' b)^2] = \arg \min_{b \in \mathbb{R}^d} E[\Delta_\tau^2(X_i, b)].$$

In other words, $\tilde{\beta}_\tau$ is the slope of the linear regression of $Q_\tau(Y_i|X_i)$ on X_i , weighted by the histogram of X_i . In contrast with quantile regression, which requires only one pass through the data, MD relies on the ability to estimate $Q_\tau(Y_i|X_i)$ consistently in a nonparametric first step.

Figure 1 plots MD fitted values with a dashed line. The quantile regression and MD lines are close, but they are not identical because of the weighting by $w_\tau(X_i, \beta_\tau)$ in the quantile regression fit. This weighting accentuates the quality of the fit at values of X_i where Y_i is more densely distributed near the CQF. Panels D-F in Figure 7.1.1 plot the overall quantile weights, $w_\tau(X_i, \beta_\tau) \cdot \pi(X_i)$ against X_i . The panels also show estimates of the $w_\tau(X_i, \beta_\tau)$, labeled "importance weights," and their density approximations, $1/2 \cdot f_Y(Q_\tau(Y_i|X_i)|X_i)$. The importance weights and the density weights are similar and fairly flat. The overall weighting function looks a lot like the schooling histogram, and therefore places the highest weight on 12 and 16 years of schooling.

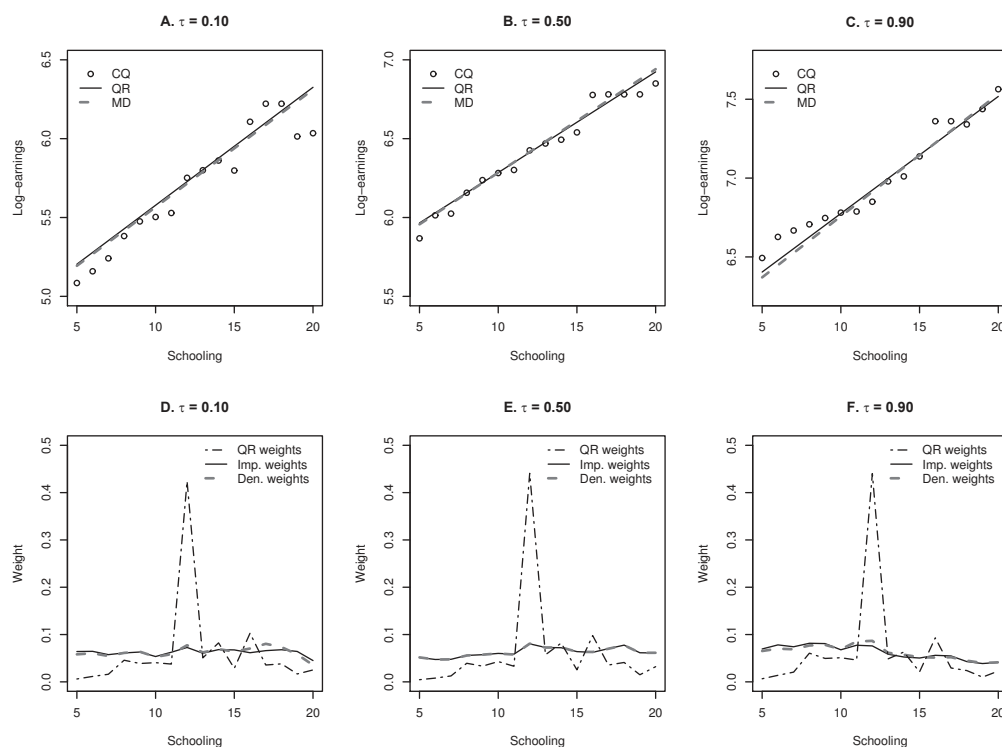


Figure 7.1.1: The quantile regression approximation property (adapted from Angrist, Chernozhukov, and Fernandez-Val, 2006). The figure shows alternative estimates of the conditional quantile function of log wages given highest grade completed using 1980 Census data, along with the implied weighting function. Panels A-C report nonparametric (CQ), quantile regression (QR) and minimum distance estimates (MD) for $\tau = .1, .5, .9$. Panels D-F show the corresponding weighting functions for QR and MD, as explained in the text.

7.1.3 Tricky Points

The language of conditional quantiles is tricky. Sometimes we talk about "quantile regression coefficients at the median," or "effects on those at the lower decile." But it's important to remember that quantile coefficients tell us about effects on *distributions* and not on *individuals*. If we discover, for example, that a training program raises the lower decile of the wage distribution, this does not necessarily mean that someone who would have been poor (i.e. at the lower decile without training) is now less poor. It only means that those who are poor in the regime with training are less poor than the poor would be in a regime without training.

The distinction between making a given set of poor people richer and changing what it means to be poor is subtle. This distinction has to do with whether we think an intervention preserves an individual's rank in the wage (or other dependent variable) distribution. If an intervention is rank-preserving, then an increase in the lower decile indeed makes those who would have been poor richer since rank preservations means relative status is unchanged. Otherwise, we can only say that the poor - defined as the group in the bottom 10 percent of the wage distribution, whoever they may be - are better off. We elaborate on this point briefly in Section 7.2, below.

A second tricky point is the transition from conditional quantiles to marginal quantiles. A link from conditional to marginal quantiles allows us to investigate the impact of changes in quantile regression coefficients on overall inequality. Suppose, for example, that quantile coefficients fan out even further with schooling, beyond what's observed in the 2000 Census. What does this imply for the ratio of upper-decile to lower-decile wages? Alternately, we can ask: how much of the overall increase in inequality (say, as measured by the ratio of upper- to lower-deciles) is explained by the fanning out of quantile regression coefficients? These sorts of questions turn out to be surprisingly difficult to answer. The difficulty has to do with the fact that all conditional quantiles are needed to pin down a particular marginal quantile (Machado and Mata, 2005). In particular, $Q_\tau(Y_i|X_i) = X_i'\beta_\tau$ does *not* imply $Q_\tau(Y_i) = Q_\tau(X_i)'\beta_\tau$. This contrast this with the much more tractable expectations operator, where if $E(Y_i|X_i) = X_i'\beta$, then by iterating expectations, we have $E(Y_i) = E(X_i)'\beta$.

Extracting marginal quantiles*

To show the link between conditional quantiles and marginal distributions more formally, suppose the CQF is indeed linear, so that $Q_\tau(Y_i|X_i) = X_i'\beta_\tau$. Let $F_Y(y|X_i) \equiv P[Y_i < y|X_i]$ with marginal distribution $F_Y(y) = P[Y_i < y]$. By definition of a conditional quantile,

$$\int_0^1 1[F_Y^{-1}(\tau|X_i) < y]d\tau = F_Y(y|X_i). \quad (7.1.9)$$

In other words, the proportion of the population below y conditional on X_i is the same as the proportion of conditional quantiles that are below y .⁴ Substituting for the CQF inside the integral,

$$F_Y(y|X_i) = \int_0^1 1[X'_i\beta_\tau < y]d\tau.$$

Next, we use the CDF of X_i , $F_X(x)$, to integrate and get the marginal distribution function, $F_Y(y)$:

$$F_Y(y) = \int \int_0^1 1[X'_i\beta_\tau < y]d\tau dF_X(x). \quad (7.1.10)$$

Finally, marginal quantiles, say, $Q_\tau(y_i)$ for $\tau \in (0, 1)$, come from inverting $F_Y(y)$:

$$Q_\tau(y_i) = \inf \{y : F_Y(y) \geq \tau\}.$$

An estimator of the marginal distribution replaces integrals with sums in (7.1.10), where the integral over quantiles comes from quantile regression estimates at, say, every .01 quantile. In a sample of size n , this is:

$$\hat{F}_Y(y) = n^{-1} \sum_i (1/100) \sum_{\tau=0}^{\tau=1} 1[X'_i\hat{\beta}_\tau < y].$$

The corresponding marginal quantile estimator inverts $\hat{F}_Y(y)$.

There are a number of difficulties with this approach in practice. For one thing, you have to estimate lots of quantile regressions. Another is that the distribution theory is messy (though not insurmountable; see, e.g., Melly, 2005). Simplifying the conditional-to-marginal quantile transition is an area of active research. Gosling, Machin, and Meghir (2000) and Machado and Mata (2005) are among the first empirical studies to go from conditional to marginal quantiles. When the variable of primary interest in a quantile regression model is a dummy variable and the other regressors are seen as controls, a propensity-score type weighting scheme can be used to produce the difference in quantiles conditional on the dummy. See Firpo (2007) for the exogenous case and Frolich and Melly (2007) for a marginalization scheme that works for quantile treatment effects of the sort discussed in the next section.

7.2 Quantile Treatment Effects

The \$42,000 question regarding any set of regression estimates is whether they have a causal interpretation. This is no less true for quantile regression than Ordinary Least Squares. Suppose we are interested in estimating the effect of a training program on earnings. OLS regression estimates measure the impact of the

⁴For example, if y is the conditional median, then $F_Y(y|X_i) = .5$ and half of all conditional quantiles are below y . The relation (7.1.9) can be proved formally using the change of variables formula.

program on average earnings while quantile regression estimates can be used to measure the impact of the program on median earnings. In both cases, we must worry about whether the estimated program effects are contaminated by omitted variables bias.

Here too, omitted variables problems can be solved using instrumental variables, though IV methods for quantile models are a relatively new development, and are not yet as flexible as conventional 2SLS. We discuss an approach that captures the causal effect of a binary variable on quantiles (i.e., a treatment effect) using a binary instrument. The Quantile Treatment Effects (QTE) estimator, introduced in Abadie, Angrist, and Imbens (2002), relies on essentially the same assumptions as the LATE framework for average causal effects. The result is an Abadie-type weighting estimator of the causal effect of treatment on quantiles for compliers.⁵

Our discussion of the QTE estimator is based on an additive model for conditional quantiles, so that a single treatment effect is estimated. The resulting estimator simplifies to Koenker and Bassett (1978) linear quantile regression when there is no instrumenting. The relationship between QTE and quantile regression is therefore analogous to that between conventional 2SLS and OLS when the regressor of interest is a dummy.

The parameters of interest are defined as follows. For $\tau \in (0, 1)$, we assume there exist $\alpha_\tau \in \mathbb{R}$ and $\beta_\tau \in \mathbb{R}^r$ such that

$$Q_\tau(Y_i|X_i, D_i, D_{1i} > D_{0i}) = \alpha_\tau D_i + X_i' \beta_\tau, \quad (7.2.1)$$

where $Q_\tau(Y_i|X_i, D_i, D_{1i} > D_{0i})$ denotes the τ -quantile of Y_i given X_i and D_i for compliers. Thus, α_τ and β_τ are quantile regression coefficients for compliers.

Recall that D_i is independent of potential outcomes conditional on X_i and $D_{1i} > D_{0i}$, as we discussed in (4.5.2). The parameter α_τ in this model therefore gives the difference in the conditional-on- X_i quantiles of Y_{1i} and Y_{0i} for compliers. In other words,

$$Q_\tau(Y_{1i}|X_i, D_{1i} > D_{0i}) - Q_\tau(Y_{0i}|X_i, D_{1i} > D_{0i}) = \alpha_\tau \quad (7.2.2)$$

This tells us, for example, whether a training program changed the conditional median or lower decile of earnings for compliers. Note that the parameter α_τ does not tell us whether treatment changed the quantiles of the unconditional distributions of Y_{1i} and Y_{0i} . For that, we have to integrate families of quantile regression results using procedures like the one described in Section 7.1.3.

It also bears emphasizing that α_τ is not the conditional quantile of the individual treatment effects, $(Y_{1i} - Y_{0i})$. You might want to know, for example, whether the median treatment effect is positive. Unfortunately, questions like this are very hard to answer without making the assumptions usually invoked for causal inference.⁶ Even a randomized trial with perfect compliance fails to reveal the distribution of

⁵For an alternative approach, see Chernozhukov and Hansen (2005), which allows for regressors of any type (i.e., not just dummies), but invokes a rank-invariance assumption that is unnecessary in the QTE framework.

⁶See, for example, Heckman, Smith, and Clements (1997).

$(Y_{1i} - Y_{0i})$. This does not matter for average treatment effects since the mean of a difference is the difference in means. But all other features of the distribution of $Y_{1i} - Y_{0i}$ are hidden because we never get to see both Y_{1i} and Y_{0i} for any one person. The good news for applied econometricians is that the difference in marginal distributions, (7.2.2), is usually more important than the distribution of treatment effects because comparisons of aggregate economic welfare typically require only the marginal distributions of Y_{1i} and Y_{0i} and not the distribution of their difference (see, e.g., Atkinson (1970), for the traditional view). This point can be made by example without reference to quantiles. When evaluating an employment program, we are inclined to view the program favorably if it increases overall employment rates. In other words, we are happy if the average Y_{1i} is higher than the average Y_{0i} . The number of individuals who gain jobs ($Y_{1i} - Y_{0i} = 1$) or lose jobs ($Y_{1i} - Y_{0i} = 0$) seems like it should be of secondary interest since a good program will necessarily have more gainers than losers.

7.2.1 The QTE Estimator

The QTE estimator is motivated by the observation that, since the parameters of interest are quantile regression coefficients for compliers, they can (theoretically) be estimated consistently by running quantile regressions in the population of compliers. As always, however, the compliers population is not identifiable; we cannot list the compliers in a given data set. Nevertheless, as in Section 4.5.2, the relevant econometric minimand can be constructed using the Abadie Kappa theorem. Specifically,

$$(\alpha_\tau, \beta_\tau) = \arg \min_{a,b} E\{\rho_\tau(Y_i - aD_i - X_i'b) | D_{1i} > D_{0i}\} = \arg \min_{a,b} E\{\kappa_i \rho_\tau(Y_i - aD_i - X_i'b)\}, \quad (7.2.3)$$

where

$$\kappa_i = 1 - \frac{D_i(1 - Z_i)}{1 - P(Z_i = 1 | X_i)} - \frac{(1 - D_i)Z_i}{P(Z_i = 1 | X_i)},$$

as before. The QTE estimator is the sample analog of (7.2.3).

There are a number of practical issues that arise when implementing QTE. First, κ_i must be estimated and the sampling variance induced by this first-step estimation should be reflected in the relevant asymptotic distribution theory. Abadie, Angrist, and Imbens (2002) derive the limiting distribution of the sample analog of (7.2.3) when κ_i is estimated nonparametrically. In practice, however, it is easier to bootstrap the whole procedure (i.e., beginning with the construction of estimated kappas) than to use the asymptotic formulas.

Second, κ_i is negative when $D_i \neq Z_i$. The kappa-weighted quantile regression minimand is therefore non-convex and no longer has a linear programming representation. This problem can be solved by working with the following minimization problem instead:

$$\min_{a,b} E\{E[\kappa_i | Y_i, D_i, X_i] \rho_\tau(Y_i - aD_i - X_i'b)\} \quad (7.2.4)$$

This minimand is derived by iterating expectations in (7.2.3). The practical difference between (7.2.3) and (7.2.4) is that the term

$$E[\kappa_i|Y_i, D_i, X_i] = P[D_{1i} > D_{0i}|Y_i, D_i, X_i]$$

is a probability and therefore between zero and one.⁷ A further simplification comes from the fact that

$$E[\kappa_i|Y_i, D_i, X_i] = 1 - \frac{D_i(1 - E[Z_i|Y_i, D_i = 1, X_i])}{1 - P(Z_i = 1|X_i)} - \frac{(1 - D_i)E[Z_i|Y_i, D_i = 0, X_i]}{P(Z_i = 1|X_i)}. \quad (7.2.5)$$

Angrist (2001) uses this to implement QTE via a Probit first step to estimate $E[Z_i|Y_i, D_i, X_i]$ separately in the $D_i = 0$ and $D_i = 1$ subsamples, constructing $E[\kappa_i|Y_i, D_i, X_i]$ using (7.2.5), and then trimming any of the resulting estimates of $E[\kappa_i|Y_i, D_i, X_i]$ that are outside the unit interval. The resulting first-step estimates of $E[\kappa_i|Y_i, D_i, X_i]$ can simply be plugged in as weights when constructing quantile regression estimates in a second step using Stata's *qreg* command.⁸

Estimates of the Effect of Training on the Quantiles of Trainee Earnings

The Job Training Partnership Act was a large federal program that provided subsidized training to disadvantaged American workers in the 1980s. JTPA services were delivered at 649 sites, also called Service Delivery Areas (SDAs), located throughout the country. The original study of the labor-market impact of JTPA services was based on 15,981 people for whom continuous data on earnings (from either State unemployment insurance (UI) records or two follow-up surveys) were available for at least 30 months after random assignment.⁹ There are 6,102 adult women with 30-month earnings data and 5,102 adult men with 30-month earnings data.

In our notation, Y_i is 30-month earnings, D_i indicates enrollment for JTPA services, and Z_i indicates the randomly assigned *offer* of JTPA services. A key feature of most social experiments, as with many randomized trials of new drugs and therapies, is that some participants decline the intervention being offered. In the JTPA, those offered services were not compelled to participate in training. Consequently, although the offer of subsidized training was randomly assigned, only about 60 percent of those offered training actually received JTPA services. Treatment received is therefore partly self-selected and likely to be correlated with potential outcomes. On the other hand, the randomized offer of training provides a good instrument for training received since the two are obviously correlated and the offer of treatment is independent of potential

⁷Intuitively, this is because κ_i "finds compliers". A formal statement of this result appears in Abadie, Angrist, and Imbens (2002; Lemma 3.2).

⁸Step-by-step, it goes like this:

1. Probit Z_i on Y_i and X_i separately in the $D_i = 0$ and $D_i = 1$ subsamples. Save these fitted values.
2. Probit Z_i on X_i in the whole sample. Save these fitted values.
3. Construct $E[\kappa_i|Y_i, D_i, X_i]$ by plugging the two sets of fitted values into (7.2.5). Set anything less than zero to zero and anything greater than one to one.
4. Use these kappas to weight quantile regressions.
5. Bootstrap this whole procedure to construct standard errors.

⁹See Bloom *et al* (1997).

outcomes. Moreover, because of the very low percentage of individuals receiving JTPA services in the control group (less than 2 percent), effects for compliers in this case can be interpreted as effects on those who were treated (there are few always-takers).

Since training offers were randomized in the National JTPA Study, covariates (X_i) are not required to consistently estimate effects on compliers. Even in experiments like this, however, it's customary to control for covariates to correct for chance associations between treatment status and applicant characteristics and to increase precision (see Chapter 2). The covariates used here are baseline measures from the JTPA intake process. They include dummies for black and Hispanic applicants, a dummy for high-school graduates (including GED holders), dummies for married applicants, 5 age-group dummies, and a dummy for whether the applicant worked at least 12 weeks in the year preceding random assignment. Also included are dummies for the original recommended service strategy (classroom, on-the-job training (OJT), job search assistance (JSA), other) and a dummy for whether earnings data are from the second follow-up survey. Since these covariates mostly summarize coarse demographics, we can think of the quantile analysis as telling us how the JTPA experiment affected the earnings distribution within demographic groups.

As a benchmark, OLS and conventional instrumental variables (2SLS) estimates of the impact of training are reported in the first column of Table 7.2.1. The OLS training coefficient is a precisely estimated \$3,754. This is the coefficient on D_i in a regression of Y_i on D_i and X_i . These estimates ignore the fact that trainees are self-selected. The 2SLS estimates in Table 7.2.1 use the randomized offer of treatment Z_i as an instrument for D_i . The 2SLS estimate for men is \$1,593 with a standard error of \$895, less than half the size of the corresponding OLS estimate.

Quantile regression estimates show that the gap in quantiles by trainee status is much larger (in proportionate terms) below the median than above it. This can be seen in the right-hand columns of Table 7.2.1, which reports quantile regression estimates for the .15, .25, .5, .75, and .85 quantiles. Specifically, the .85 quantile of trainee earnings is about 13 percent higher than the corresponding quantile for non-trainees, while the .15 quantile is 136 percent higher. Like the OLS estimates in the table, these quantile regression coefficients do not necessarily have a causal interpretation. Rather they provide a descriptive comparison of the earnings distributions of trainees and non-trainees.

QTE estimates of the effect of training on median earnings are similar in magnitude though less precise than the benchmark 2SLS estimates. On the other hand, the QTE estimates show a pattern very different from the quantile regression estimates, with no evidence of an impact on the .15 or .25 quantile. The estimates at low quantiles are substantially smaller than the corresponding quantile regression estimates, and they are small in absolute terms. For example, the QTE estimate (standard error) of the effect on the .15 quantile is \$121 (475), while the corresponding quantile regression estimate is \$1,187 (205). Similarly, the QTE estimate (standard error) of the effect on the .25 quantile for men is \$702 (670), while the corresponding quantile regression estimate is \$2,510 (356). Unlike the results at low quantiles, however, the QTE estimates

of effects on male earnings above the median are large and statistically significant (though still smaller than the corresponding quantile regression estimates).

The result that JTPA training for adult men did not raise the lower quantiles of their earnings is the most interesting finding arising from this analysis. This suggests that the quantile regression estimates in the top half of Table 7.2.1 are contaminated by positive selection bias. One response to this finding might be that few JTPA applicants were very well off, so that distributional effects within applicants are of less concern than the fact that the program helped many applicants overall. However, the upper quantiles of earnings were reasonably high for adults who participated in the National JTPA Study. Increasing earnings in this upper tail is therefore unlikely to have been a high priority.

Table 7.2.1: Quantile regression estimates and quantile treatment effects from the JTPA experiment

A. OLS and Quantile Regression Estimates						
	OLS	Quantile				
		0.15	0.25	0.50	0.75	0.85
Training	3,754 (536)	1,187 (205)	2,510 (356)	4,420 (651)	4,678 (937)	4,806 (1,055)
% Impact of Training	21.20	135.56	75.20	34.50	17.24	13.43
High school or GED	4,015 (571)	339 (186)	1,280 (305)	3,665 (618)	6,045 (1,029)	6,224 (1,170)
Black	-2,354 (626)	-134 (194)	-500 (324)	-2,084 (684)	-3,576 (1,087)	-3,609 (1,331)
Hispanic	251 (883)	91 (315)	278 (512)	925 (1,066)	-877 (1,769)	-85 (2,047)
Married	6,546 (629)	587 (222)	1,964 (427)	7,113 (839)	10,073 (1,046)	11,062 (1,093)
Worked less than 13 weeks in past year	-6,582 (566)	-1,090 (190)	-3,097 (339)	-7,610 (665)	-9,834 (1,000)	-9,951 (1,099)
Constant	9,811 (1,541)	-216 (468)	365 (765)	6,110 (1,403)	14,874 (2,134)	21,527 (3,896)
B. 2SLS and QTE Estimates						
	2SLS	Quantile				
		0.15	0.25	0.50	0.75	0.85
Training	1,593 (895)	121 (475)	702 (670)	1,544 (1,073)	3,131 (1,376)	3,378 (1,811)
% Impact of Training	8.55	5.19	11.99	9.64	10.69	9.02
High school or GED	4,075 (573)	714 (429)	1,752 (644)	4,024 (940)	5,392 (1,441)	5,954 (1,783)
Black	-2,349 (625)	-171 (439)	-377 (626)	-2,656 (1,136)	-4,182 (1,587)	-3,523 (1,867)
Hispanic	335 (888)	328 (757)	1,476 (1,128)	1,499 (1,390)	379 (2,294)	1,023 (2,427)
Married	6,647 (627)	1,564 (596)	3,190 (865)	7,683 (1,202)	9,509 (1,430)	10,185 (1,525)
Worked less than 13 weeks in past year	-6,575 (567)	-1,932 (442)	-4,195 (664)	-7,009 (1,040)	-9,289 (1,420)	-9,078 (1,596)
Constant	10,641 (1,569)	-134 (1,116)	1,049 (1,655)	7,689 (2,361)	14,901 (3,292)	22,412 (7,655)

Notes: The table reports OLS, quantile regression, 2SLS, and QTE estimates of the effect of training on earnings (adapted from Abadie, Angrist, and Imbens (2002)). Assignment status is used as an instrument for training status in Panel B. All models include as covariates dummies for service strategy recommended and age group, and a dummy indicating data from a second follow-up survey. Robust standard errors are reported in parenthesis.